

Vector Databases

Vector databases are specialized database systems designed to store and query high-dimensional vector embeddings. Unlike traditional relational databases that handle structured data, vector databases excel at similarity search operations. They can efficiently find the most similar items to a query vector from millions or billions of candidates. Popular vector databases include ChromaDB, Pinecone, Weaviate, Milvus, and FAISS. These databases use approximate nearest neighbor (ANN) algorithms like HNSW and IVF to achieve fast search speeds while maintaining accuracy. Vector databases are essential for semantic search, recommendation systems, and RAG applications.